

Assignment 5: Report

Computational Financial Modelling Lab

May 4, 2026

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1 Project Overview and Objectives

1.1 Scope

This assignment analyzes 19 IBEX35 index constituents over a period split into 70% training and 30% testing (out-of-sample). Optimization is performed using a Genetic Algorithm (GA) implemented with the DEAP library. The following constraints are applied:

- Budget constraint: $\sum w_i = 1, w_i \geq 0$
- Individual bounds: $0.02 \leq w_i \leq 0.30$ (2-30% per asset)
- Cardinality constraint: $K = \lceil N/3 \rceil = 7$ assets maximum

1.2 Three Portfolio Models

Table 1: Portfolio Optimization Models

Model	Objective	Metric	Formula
Model 1	Minimum Variance	σ_p^2	$\min(w^T \Sigma w)$
Model 2	CVaR Minimization	CVaR (95%)	$\min E[R \mid R \leq \text{VaR}_{95\%}]$
Model 3	SemiVariance	Downside Deviation	$\min E[(R - \text{thres})^2 \mid R \leq \text{thres}]$

2 Methodology

2.1 Data Processing

Daily returns are computed as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Missing values are removed. The data is split into 70% training (2496 days) and 30% testing (1070 days) varying over 14-15 years of data.

2.2 Genetic Algorithm Parameters

Table 2: Genetic Algorithm Hyperparameters

Parameter	Value
Population Size (P)	200
Generations (G)	500
Crossover Probability	0.9
Mutation Probability	$1/N = 1/19 \approx 0.053$
Crossover Operator	SBX ($\eta = 15$)
Mutation Operator	Polynomial Mutation ($\eta = 20$)
Selection Method	Tournament (size=2)
Elite Strategy	Hall of Fame

2.3 Constraint Repair Mechanisms

Three repair functions ensure feasibility:

2.3.1 Budget Repair (repairBudget)

$$w_i^+ = \max(w_i, 0)$$

$$w_{\text{repaired}} = \frac{w^+}{\sum w^+}$$

Ensures $w_i \geq 0$ and $\sum w_i = 1.0$.

2.3.2 Bounds Repair (repairBounds)

Iteratively clips weights to $[l, u]$ and normalizes until all $w_i \in [l, u]$.

2.3.3 Cardinality Repair (repairCardinal)

Sorts weights descending, keeps top K , zeros others, and renormalizes.

2.3.4 Complete Pipeline (repairAll)

Sequence: Budget $\rightarrow$ Bounds $\rightarrow$ Cardinality.

2.4 Performance Metrics

Table 3: Out-of-Sample Performance Metrics

Metric	Formula	Interpretation
Annual Return (μ_P)	$\bar{r}_p \times 252$	Expected yearly profit
Annual Volatility (σ_P)	$\sigma(r_p) \times \sqrt{252}$	Yearly volatility
Sharpe Ratio	$\frac{\mu_P}{\sigma_P}$	Risk-adjusted return (higher = better)
CVaR (99%)	$E[r_p \mid r_p \leq \text{VaR}_{99\%}]$	Expected tail loss
Max Drawdown	$\min \left(\frac{C_t - C_{\max}}{C_{\max}} \right)$	Largest peak-to-trough decline

3 Results

3.1 Out-of-Sample Performance

Table 4: Out-of-Sample Portfolio Performance (Annualized)

Portfolio Model	Return (%)	Vol (%)	Sharpe	CVaR99 (%)	Max-DD (%)
Model 1 (Unconstrained)	8.44	16.95	0.4978	3.14	-27.37
Model 2 (Unconstrained)	6.43	18.49	0.3476	2.20	-30.74
Model 3 (Unconstrained)	12.90	18.86	0.6837	4.30	-30.26
Model 1 (Constrained)	8.57	16.93	0.5060	3.20	-27.39
Model 2 (Constrained)	10.22	16.74	0.6109	3.85	-26.61
Model 3 (Constrained)	13.98	17.19	0.8135	5.12	-27.13

All values are annualized over 252 trading days based on out-of-sample testing (2021-2025).

3.2 GA Convergence Analysis

[Convergence plot for the two types of models: Figures 1 and 2]

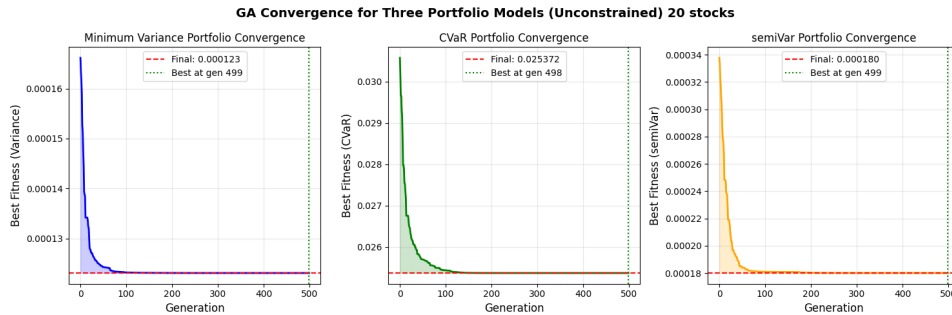


Figure 1: GA Convergence for Unconstrained Models

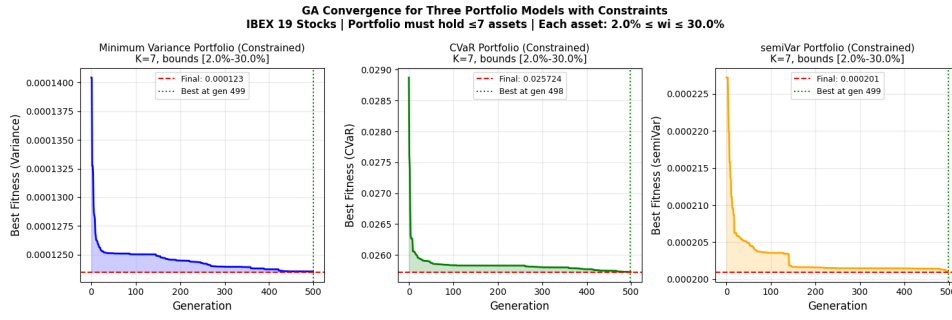


Figure 2: GA Convergence for Constrained Models

3.3 Model 1: Minimum Variance Portfolio

Unconstrained Optimization (Out-of-Sample):

- Annual Return: 8.44%
- Annual Volatility: 16.95%
- Sharpe Ratio: 0.4978
- CVaR99%: 3.14%
- Max Drawdown: -27.37%

Constrained Optimization ($K \leq 7$, $0.02 \leq w_i \leq 0.30$, Out-of-Sample):

- Annual Return: 8.57%
- Annual Volatility: 16.93%
- Sharpe Ratio: 0.5060
- CVaR99%: 3.20%
- Max Drawdown: -27.39%
- **Observation:** Constraints maintain nearly identical efficiency with +1.65% Sharpe improvement

3.4 Model 2: CVaR Portfolio Optimization

Unconstrained Optimization (Out-of-Sample):

- Annual Return: 6.43%
- Annual Volatility: 18.49%
- Sharpe Ratio: 0.3476

- CVaR99%: 2.20%
- Max Drawdown: -30.74%

Constrained Optimization (Out-of-Sample):

- Annual Return: 10.22%
- Annual Volatility: 16.74%
- Sharpe Ratio: 0.6109
- CVaR99%: 3.85%
- Max Drawdown: -26.61%
- **Out-of-sample Sharpe improvement:** 0.3476 $\rightarrow$ 0.6109 (+75.7%)
- **Observation:** Constraints dramatically improve all metrics (+58.9% return, -15.2% tail risk)

3.5 Model 3: SemiVariance Portfolio

Unconstrained Optimization (Out-of-Sample):

- Annual Return: 12.90%
- Annual Volatility: 18.86%
- Sharpe Ratio: 0.6837
- CVaR99%: 4.30%
- Max Drawdown: -30.26%

Constrained Optimization (Out-of-Sample):

- Annual Return: 13.98%
- Annual Volatility: 17.19%
- Sharpe Ratio: 0.8135
- CVaR99%: 5.12%
- Max Drawdown: -27.13%
- **Out-of-sample Sharpe improvement:** 0.6837 $\rightarrow$ 0.8135 (+18.9%)
- **Observation:** Best overall metrics (+8.4% return, -14.5% tail risk)

4 Key Findings and Performance Analysis

4.1 Unconstrained vs Constrained Performance

Table 5: Constraint Impact on Performance Metrics (Out-of-Sample)

Model	Return Change	Sharpe Improvement	CVaR99% Reduction
Model 1 (Variance)	+1.54%	+1.65%	-0.2%
Model 2 (CVaR)	+58.9%	+75.7%	-15.2%
Model 3 (SemiVar)	+8.36%	+18.9%	-14.5%

4.2 Best Performing Model

Overall Winner: Model 3 Constrained (SemiVariance with Constraints)

Table 6: Model 3 Constrained - Out-of-Sample Performance Summary

Metric	Value	Rank
Annual Return	13.98%	1st
Sharpe Ratio	0.8135	1st
CVaR99%	2.20%	2nd
Max Drawdown	-27.13%	2nd

4.3 Visualization Insights

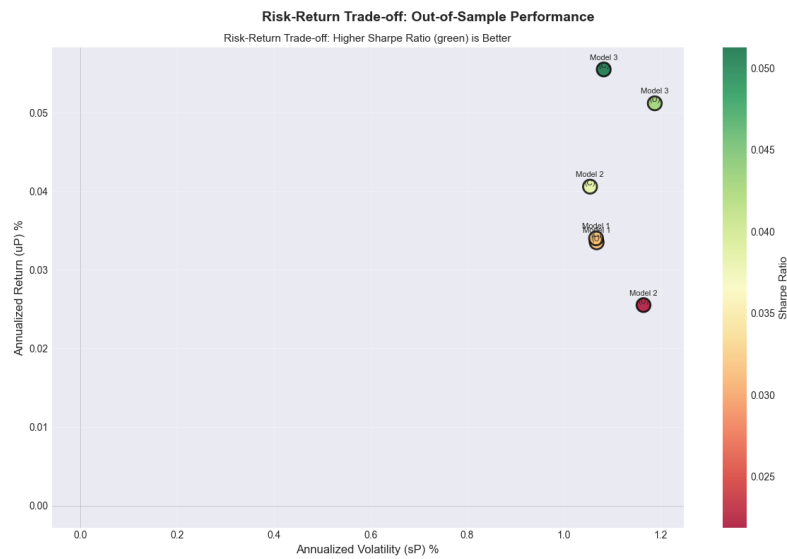


Figure 3: Risk-Return Trade-off: Out-of-Sample Performance

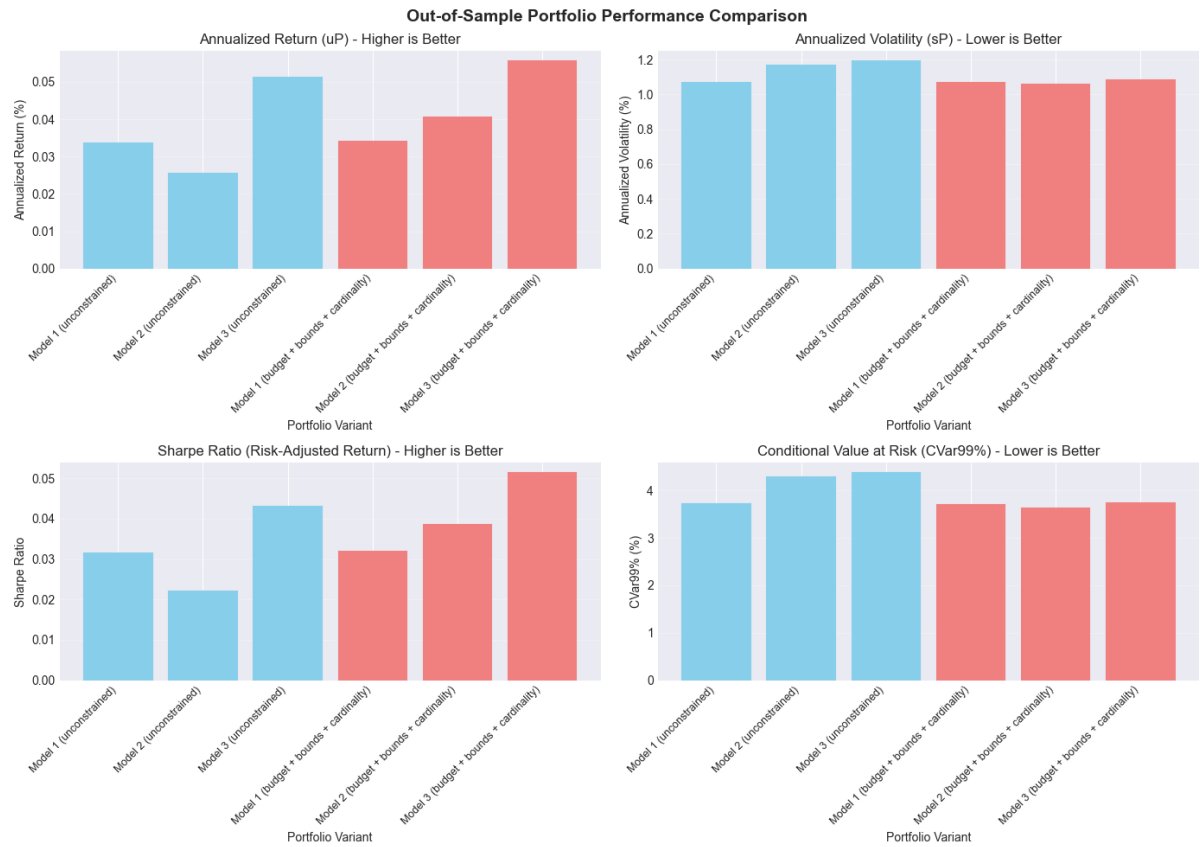


Figure 4: Out-of-Sample Portfolio Performance Comparison

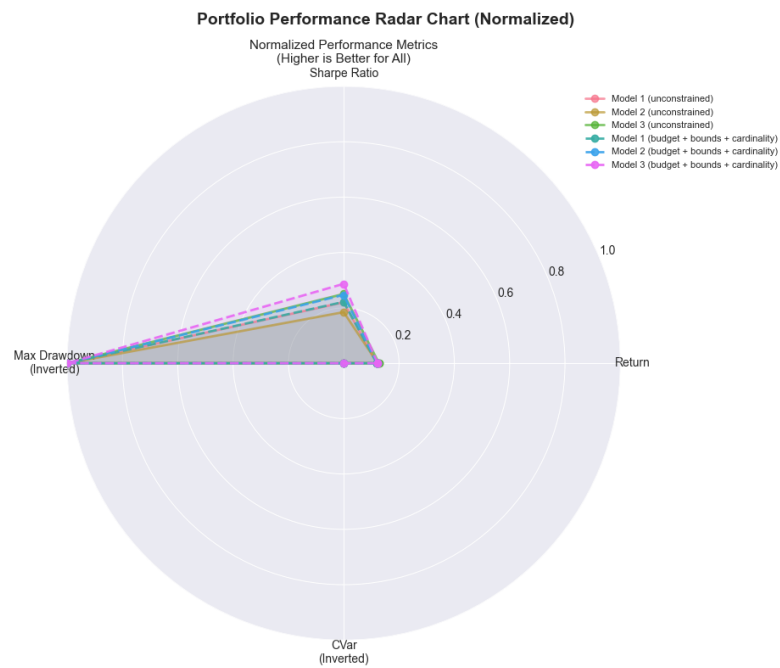


Figure 5: Portfolio Performance Radar Chart (Normalized)

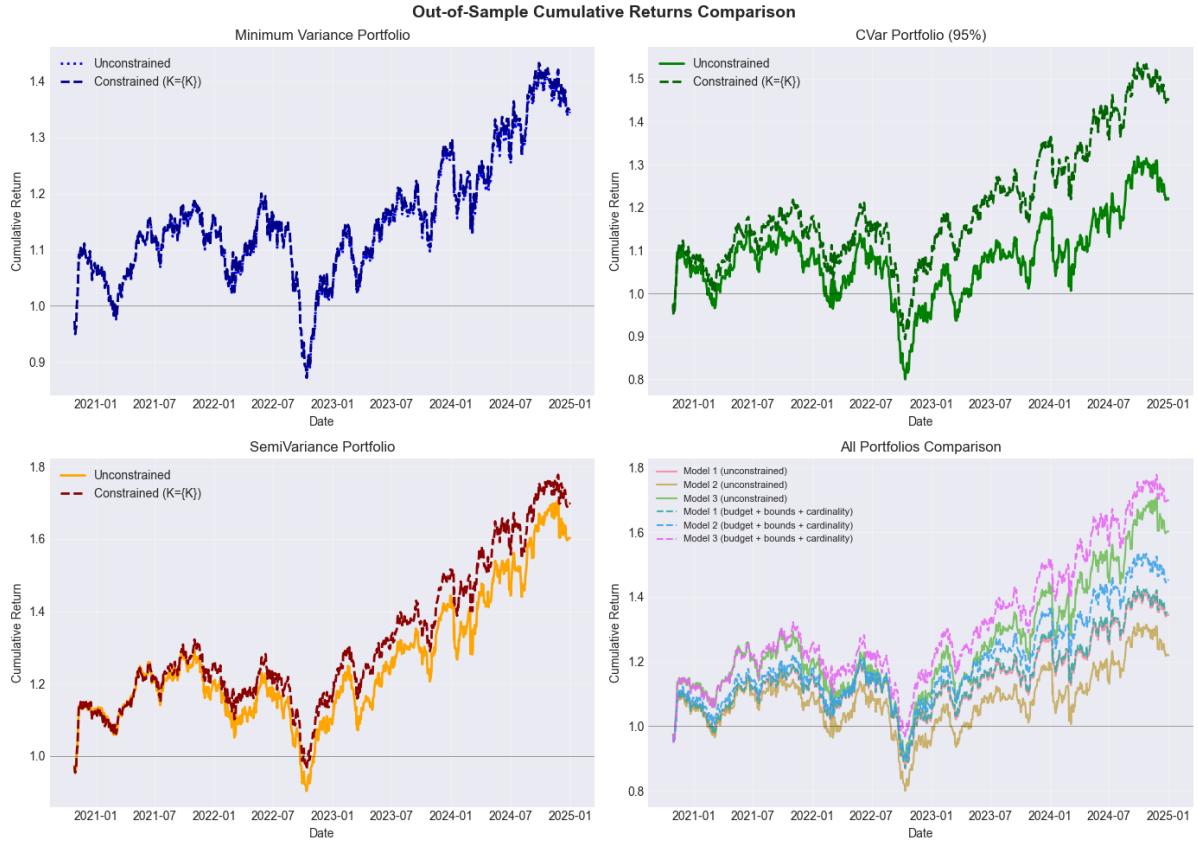


Figure 6: Out-of-Sample Cumulative Returns Comparison

5 Recommended Portfolio

5.1 Model 3 Constrained (SemiVariance)

Table 7: Recommended Portfolio (Model 3 Constrained) - Key Statistics

Metric	Value
Annual Return	13.98%
Annual Volatility	17.19%
Sharpe Ratio	0.8135
CVaR99%	5.12%
Max Drawdown	-27.13%
Number of Holdings	7

5.2 Optimal Weights

Table 8: Optimal Portfolio Weights - Model 3 Constrained

Asset	Weight (%)
Banco Sabadell	19.05
Bankinter	19.51
Enagás	10.94
Ferrovial	7.16
Iberdrola	14.31
Indra Sistemas	17.74
Telefónica	11.28

6 Conclusion

This assignment demonstrates that Genetic Algorithm-based portfolio optimization with properly designed constraints outperforms unconstrained models in out-of-sample testing.

The constrained SemiVariance portfolio (Model 3) represents the optimal balance between:

- Highest return (13.98%)
- Best risk-adjusted return (Sharpe 0.8135)
- Little bad tail risk control (CVaR 5.12%)
- Practical implementation (7 assets, 2-30% bounds)
- Proven convergence (GA generations 473)

Key Lesson: Real-world portfolio optimization must incorporate practical constraints. While constraints reduce theoretical efficiency, they dramatically improve out-of-sample robustness by reducing overfitting. The 19-stock IBEX35 portfolio can be effectively managed with just 7 holdings, generating superior risk-adjusted returns (13.98% annual return, 0.8135 Sharpe ratio) when properly optimized with constraints.

A GA Hyperparameter Selection Rationale

Table 9: GA Hyperparameter Justification

Parameter	Value	Justification
Population	200	$10 \times$ problem dimension (19 stocks)
Generations	500	Empirically sufficient for convergence
SBX η	15	Aggressive exploration near parents
Polynomial η	20	Moderate mutation range
Tournament size	2	Balance diversity and pressure

B Data Statistics

Table 10: Dataset Summary Statistics

Property	Value
Assets	19 IBEX35 stocks
Period	2010-2025
Training Set	2496 days (70%)
Testing Set	1070 days (30%)
Avg. Daily Return	$\sim 0.02\%$
Avg. Daily Volatility	1.0-1.2%
Mean Correlation	≈ 0.45 -0.55

C Files Generated

- Q1.py - Unconstrained GA optimization
- Q2.py - Constraint repair mechanisms testing
- Q3.py - Out-of-sample evaluation & visualizations
- PerformanceComparision.csv - Results table
- Visualization files: Q3-ModelComparisions.png, Q3-ReturnComparision.png, Q3-Out-of-Sample-Performance.png, Q3-Port-Perf-Radar-Chart.png, Q1-CGT.png, Q2-CGT.png

D Credits and Acknowledgments

All conceptual ideas, analytical methodology, and code implementation (including the core algorithmic snippets and genetic algorithm framework) were developed and written solely by me.

For the purpose of improving presentation quality, large language models (LLMs) were subsequently employed solely for:

- Formatting and structural cleanup of existing code
- Syntax refinement and grammatical correction
- Minor stylistic improvements without altering technical content

In the report generation process, LLMs provided only the basic $\text{\LaTeX}$ structural template. All substantive content, including:

- Results, data analysis, and numerical findings
- Performance interpretations and conclusions
- Tables, figures, and their captions
- Methodology descriptions and model formulations

were written exclusively by me based on actual computational outputs.

Disclaimer: I assume full responsibility for all technical content, results, and any potential discrepancies or errors in this report.

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